H3 Semiconductor Physics & Devices
Experiment No. L2003B PN Junction Devices
Lab Report

Cao Xizhen

Hwa Chong Institution (College Section)

Date of Experiment: April 7th, 2026

Contents

1 Introduction **2**
1.1 Overview . . . 2
1.2 Objectives . . . 2

2 Theory **2**
2.1 I-V characteristics of p-n junctions . . . 2
2.1.1 Shockley ideal diode equation . . . 2
2.1.2 Temperature dependency . . . 3
2.1.3 Thermal Generation . . . 3
2.1.4 Built-in Potential . . . 4
2.2 Light emitting diodes (LEDs) . . . 4
2.2.1 Mechanism of light emission . . . 4
2.2.2 The bandgap and light emission characteristics . . . 4
2.2.3 Spectral characteristics . . . 5
2.2.4 Factors affecting the spectrum . . . 5

3 Experimental Technique **7**
3.1 I-V characteristics . . . 7
3.2 Temperature Dependence . . . 7
3.3 Ideality factor and reverse saturation current . . . 7
3.4 Light current characteristic of an LED . . . 7
3.5 Emission spectrum of an LED . . . 7

4 Results and Discussions **7**
4.1 Finding ideality factor and reverse saturation current of diode . . . 7
4.2 I-V characteristics of diode under temperature variation . . . 7
4.3 Light-current curve . . . 7
4.4 Emission spectrum of LED . . . 10

5 Conclusion **10**

6 References **10**

1 Introduction

1.1 Overview

The p-n junction is the fundamental building block of modern semiconductor electronics, forming the basis for devices such as rectifiers, transistors, and optoelectronics. By interfacing p-type and n-type materials, a depletion region and a corresponding built-in potential are established. This barrier creates the diode's characteristic non-linear behavior: it allows exponential current flow under forward bias while acting as an insulator under reverse bias.

Beyond basic rectification, p-n junction behavior is governed by carrier statistics that are highly sensitive to thermal energy and the material's energy bandgap. This experiment explores these sensitivities by characterizing the electrical response of standard diodes across temperature gradients and observing the radiative recombination processes in Light Emitting Diodes (LEDs).

1.2 Objectives

The objective of this experiment is to:

1. obtain the I-V characteristics and parameters of a given diode,
2. observe the temperature dependency of diode I-V characteristics
3. to study and measure the light emission properties of light emitting diodes (LEDs).

2 Theory

2.1 I-V characteristics of p-n junctions

The I-V characteristic of a diode, as shown in 1 describes the relationship between the current flowing through the device (I) and the applied voltage (V) across its terminals. Unlike a resistor, which follows Ohm's Law linearly, a diode is a non-linear device that acts as a one-way valve for current [2].

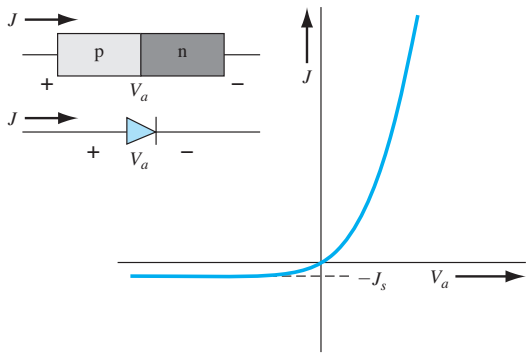

Figure 1: Ideal I-V characteristic of a p-n junction diode.

2.1.1 Shockley ideal diode equation

The behavior of the current in both forward and reverse bias can be mathematically modeled by the Shockley Diode Equation (1):

$$I = I_0(e^{\frac{qV}{nkT}} - 1) \tag{1}$$

Where I_0 is the reverse saturation current, q is the charge of an electron, k is the Boltzmann constant, T is the absolute temperature in Kelvin, and n is the ideality factor (typically 1 for Germanium and 2 for Silicon).

The I-V curve can be generally divided into three regions: forward bias ($V > 0$), reverse bias ($V < 0$), and breakdown. This experiment will only focus on the forward bias region, where the applied potential reduces the width of the depletion region, allowing majority carriers to diffuse across the junction. In this region, the exponential term in Equation 1 dominates, and the "-1" can be neglected for voltages V¿3kT/q.

2.1.2 Temperature dependency

The performance of semiconductor devices is highly sensitive to thermal variations. In a p-n junction, temperature affects the I-V characteristic primarily through the intrinsic carrier concentration (n_i), which in turn dictates the reverse saturation current (I_0).

To understand how the forward voltage V changes with temperature T at a constant current I, we begin by rearranging the simplified Shockley equation (ignoring the '-1'):

$$V = \frac{nkT}{q} \ln\left(\frac{I}{I_0}\right) \tag{2}$$

The reverse saturation current I_0 is strongly dependent on temperature because it is proportional to n_i. The relationship is given by:

$$I_0 \propto T^3 \exp\left(-\frac{E_g}{kT}\right) \tag{3}$$

Where E_g is the bandgap energy (approx. 1.12 eV for Silicon). Taking the derivative of the voltage expression with respect to temperature ($\frac{\delta V}{\delta T}$) while holding I constant involves accounting for both the linear T in the thermal voltage and the exponential T dependency within I_0. The resulting expression for the temperature coefficient is:

$$\frac{dV}{dT} = \frac{V - (E_g/q)}{T} - \frac{3nk}{q} \tag{4}$$

For a standard Silicon diode operating at room temperature with $E_g/q \approx 1.12V$, the term $(V - E_g/q)$ is negative. This yields the well-known approximation:

$$\frac{\delta V}{\delta T}|_{I=constant} \approx -2.4 \text{ mV}/^\circ\text{C} \tag{5}$$

This negative temperature coefficient indicates that as the diode heats up, the potential barrier decreases, requiring less applied voltage to maintain the same level of current.

The temperature sensitivity of a p-n junction is fundamentally rooted in the thermal generation of charge carriers. While thermal voltage increases linearly with temperature, its effect is overshadowed by the exponential growth of the reverse saturation current I_0.

2.1.3 Thermal Generation

The reverse saturation current is defined by the minority carrier concentrations and their respective diffusion constants. Mathematically, I_0 is proportional to the square of the intrinsic carrier concentration (n_i^2):

$$I_0 \propto n_i^2 = BT^3 \exp\left(-\frac{E_g}{kT}\right) \tag{6}$$

where B is a material-specific constant and E_g is the energy bandgap. As temperature increases, more electrons gain sufficient thermal energy to overcome the bandgap, leading to an exponential increase in n_i. Because I_0 represents the "leakage" of minority carriers, an increase in T effectively lowers the barrier for carrier injection under forward bias.

For a constant forward current I, an increase in temperature allows that current to be achieved at a lower applied voltage V. This phenomenon results in a negative temperature coefficient for the forward voltage. By differentiating the diode equation, we find that for Silicon, the voltage required to maintain a specific current decreases by approximately 2.4 mV for every 1°C increase in temperature.

2.1.4 Built-in Potential

Additionally, the built-in potential V_{bi} of the junction also decreases with temperature:

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) \tag{7}$$

As n_i rises with temperature, the ratio inside the logarithm decreases faster than the linear T term increases. This reduction in the built-in barrier is the primary reason the I-V curve shifts toward the origin (left) as the device heats up.

2.2 Light emitting diodes (LEDs)

A Light Emitting Diode (LED) is a semiconductor p-n junction that converts electrical energy directly into light through a process known as electroluminescence. Unlike traditional light sources that rely on heat, LEDs are "cold" light sources, making them significantly more efficient.

2.2.1 Mechanism of light emission

When a forward-bias voltage is applied to the LED, electrons from the n-side and holes from the p-side are pushed toward the active region of the junction [3].

1. Recombination: Electrons and holes meet and recombine.
2. Spontaneous Emission: As an electron moves from the higher-energy conduction band into a lower-energy hole in the valence band, it must release its excess energy.
3. Photon Release: In direct bandgap semiconductors, this energy is released as a photon. This process is called spontaneous emission because it occurs naturally without an external light trigger.

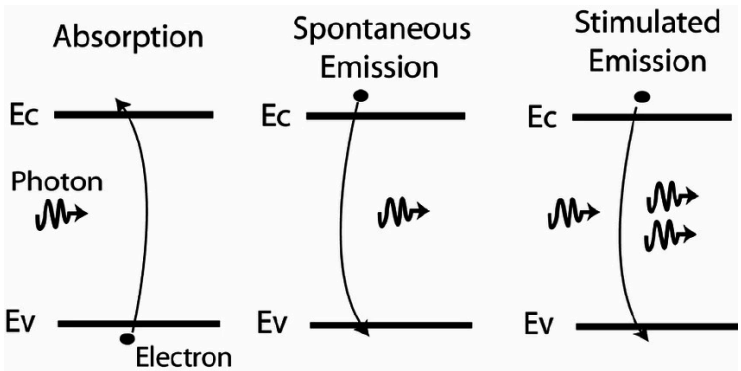

Figure 2: Illustration of absorption, spontaneous emission, and simulated emission

2.2.2 The bandgap and light emission characteristics

The energy of the emitted photon (E) is approximately equal to the bandgap energy (E_g) of the semiconductor material. This relationship is defined by the equation:

$$E = E_g = \frac{hc}{\lambda} \tag{8}$$

Where h is Planck's constant, c is the speed of light, and λ is the peak wavelength.

Because E_g is an intrinsic property of the material, the peak wavelength, and thus the specific colour of the LED, is detemined by the choice of semiconductor material.

Additionally, while all semiconductors have a bandgap, not all are suitable for efficient light emission. This is determined by the alignment of the conduction band minimum and the valence band maximum in k-space (momentum space).

1. **Direct Bandgap:** The conduction band minimum and valence band maximum align at the same momentum ($k = 0$). This allows electrons to recombine with holes and release energy directly as a **photon** without a change in momentum.

2. **Indirect Bandgap:** The band extrema are offset in momentum space. Recombination requires a third party—a phonon (lattice vibration)—to conserve momentum. This three-body interaction is inefficient, typically dissipating energy as **heat** rather than light.

2.2.3 Spectral characteristics

LEDs are often described as monochromatic, but they differ significantly from other sources:

1. LED vs. Incandescent bulbs: Incandescent bulbs use blackbody radiation, emitting a broad "white" spectrum that includes infrared (heat). LEDs use a narrow band spectrum, concentrating their energy into a specific colour, which prevents energy waste.

2. Spectral Bandwidth: While narrow, LEDs are not perfectly monochromatic like lasers. They have a finite spectral linewidth (typically 20nm-50nm) due to the thermal distribution of carriers within the energy bands.

Table 2.2.3 shows some examples of semiconductor materials used for optics fabrication [1].

Table 1: Semiconductor Materials and their Optical Characteristics

Semiconductor Name	**Composition**	**Wavelength (nm)**	**colour**	**Bandgap (eV)**
Gallium Nitride	AlGaInN	<400	Ultraviolet	3.1–4.4
Indium Gallium Nitride	InGaN	400–450	Violet	2.8–4.0
Silicon Carbide	SiC/InGaN	450–500	Blue	2.5–3.7
Gallium Phosphide	AlGaInP/GaP	500–570	Green	1.9–4.0
Gallium Arsenide Phosphide	GaAsP	570–590	Yellow	2.1–2.2
Alum. Gal. Indium Phosphide	AlGaInP	590–610	Orange	2.0–2.1
Alum. Gallium Arsenide	AlGaAs/GaAsP	610–760	Red	1.6–2.0
Gallium Arsenide	GaAs	>760	Infrared	<1.9

2.2.4 Factors affecting the spectrum

The output of an LED is not static; it can be shifted by environmental and operational factors:

1. Temperature Sensitivity: As the junction temperature increases, the semiconductor bandgap typically narrows. This causes a red-shift (longer wavelength) in the peak emission and a decrease in overall intensity.

2. Current Density: Increasing the drive current can cause a slight "blue-shift" in some materials due to band-filling effects, though this is often offset by the heat generated by the higher current.

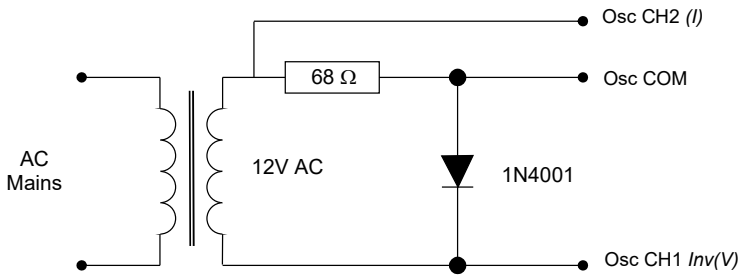

Figure 3: Sweep circuit for I-V display

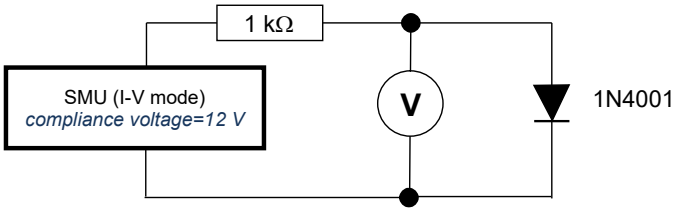

Figure 4: Circuit for diode parameter extraction

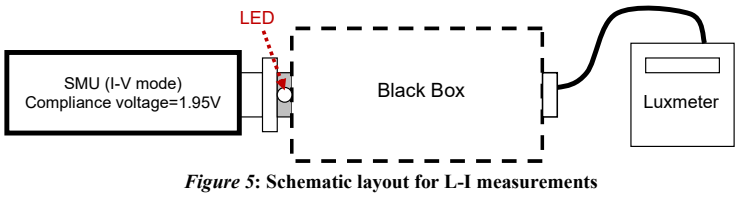

Figure 5: Schematic layout for L-I measurements

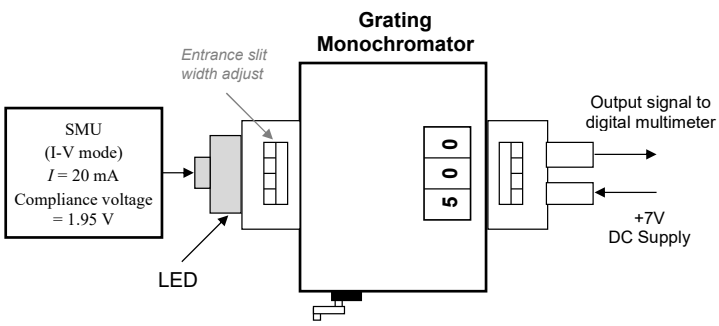

Figure 6: Measurement of emission spectrum using a grating spectrometer

3 Experimental Technique

3.1 I-V characteristics

3.2 Temperature Dependence

3.3 Ideality factor and reverse saturation current

3.4 Light current characteristic of an LED

3.5 Emission spectrum of an LED

4 Results and Discussions

4.1 Finding ideality factor and reverse saturation current of diode

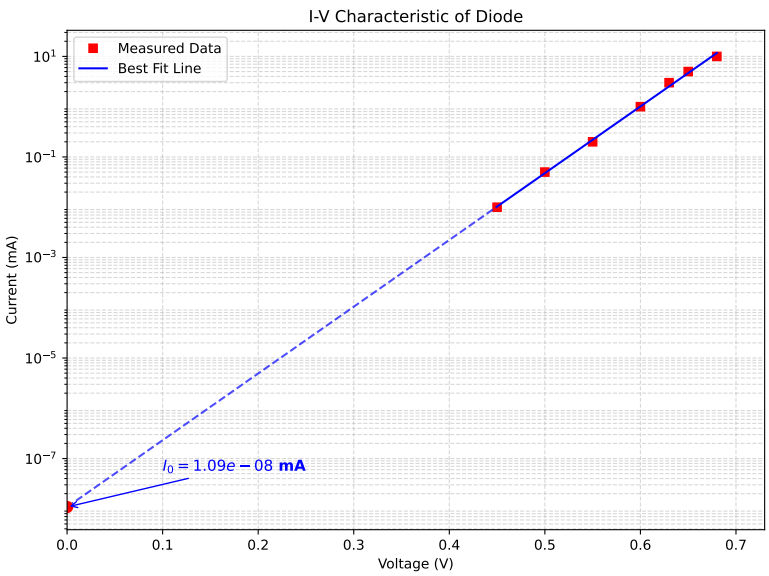

Figure 7: $\ln I$ (A) plotted against voltage (V)

4.2 I-V characteristics of diode under temperature variation

1. plot observed results sketch
2. plot ideal shockley diode equation compared to observed values

explain the disparity

4.3 Light-current curve

The data was able to be fitted to a straight line, supporting the linear relationship between current and light intensity of diode.

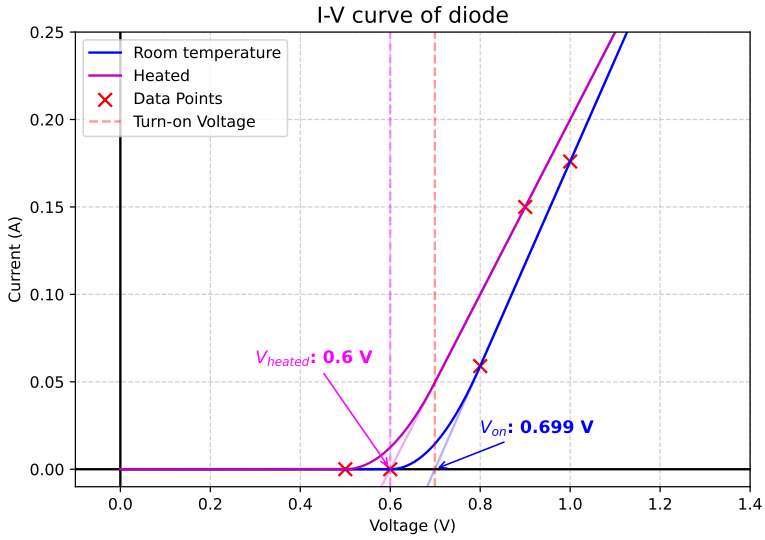

Figure 8: Sketch of IV curve of diode, at room temperature and with heating

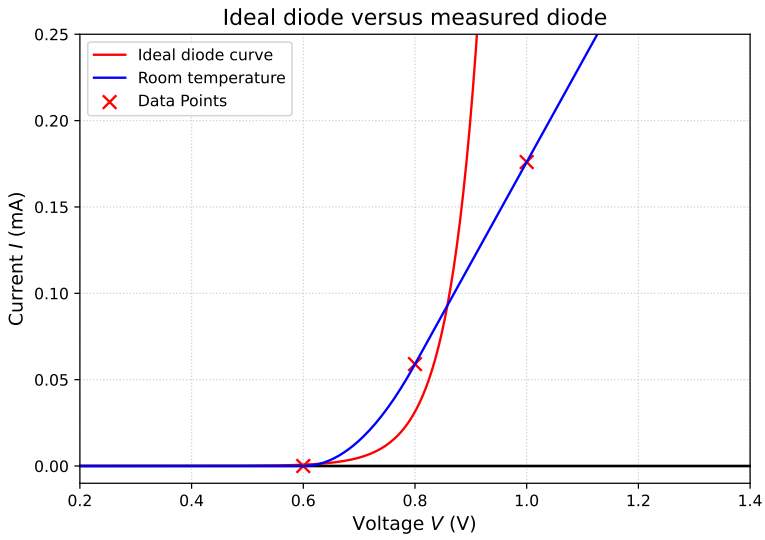

Figure 9: Ideal shockley diode I-V characteristics plotted with measured diode I-V characteristics at room temperature

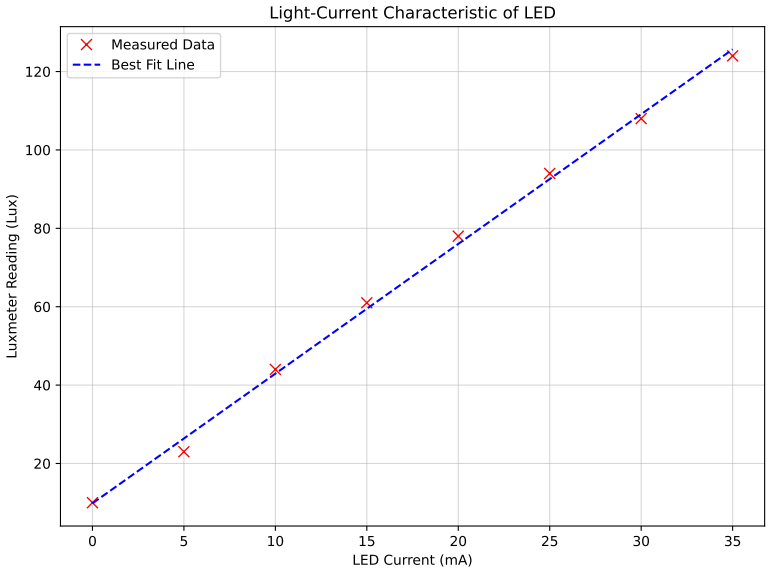

Figure 10: Light intensity plotted against current through the LED.

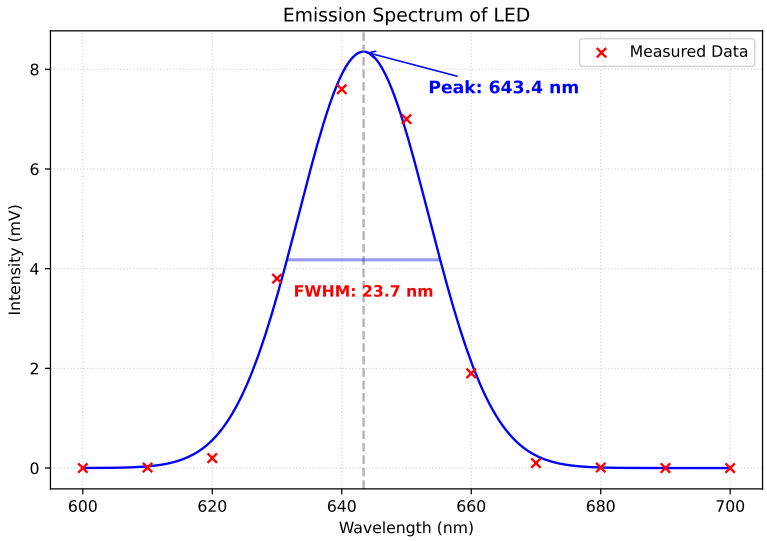

Figure 11: Light intensity plotted against emission wavelength of LED.

4.4 Emission spectrum of LED

5 Conclusion

6 References

References

[1] Detlef Klimm. "Electronic materials with a wide band gap: recent developments". In: *IUCrJ* 1.5 (Sept. 2014), pp. 281–290. DOI: 10.1107/S2052252514017229. URL: https://doi.org/10.1107/S2052252514017229.

[2] D. Neamen. *Semiconductor Physics And Devices*. McGraw-Hill, 2012. ISBN: 9780077418847. URL: https://books.google.com.sg/books?id=R_xKAgAAQBAJ.

[3] Evan P. Thrush. *Integrated Fluorescence Sensor*. Stanford University Dissertation. 2004. URL: https://www.researchgate.net/publication/260400226_INTEGRATED_FLUORESCENCE_SENSOR.